MIDTERM EXAMINATION

BSIT — Data Science (Spring 2026)

Total Marks: 30 | Time Allowed: 2 Hours

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Instructions

- Attempt **ALL** questions. All questions are **compulsory**.
- Write clear, well-structured answers. Use diagrams and formulas where appropriate.
- Marks are indicated next to each question.
- This exam is conceptual — no R/Python coding is required, but you may reference code logic or pseudo-code in your explanations.

Student Name: ______________________________________ | Roll No: _________________________ | Section: ______________ |

Question 1 — Foundations of AI, Data Science and Emerging Paradigms [10 Marks]

Read all parts carefully. Each part builds on foundational concepts from Lecture 101.

(a) [**2 marks**] Data Science is described as an interdisciplinary field. Explain the **Data Science Lifecycle** by listing its stages in the correct order (at least six stages). For each stage, write one sentence explaining what happens at that stage and why it is important.

(b) [**2 marks**] A company collects the following data about its customers: *Customer ID, Name, Age, Annual Income ($), Satisfaction Level (Low/Medium/High), Is-Active-Member (Yes/No), and Last Purchase Date*. Classify each variable as: **(i)** Numerical (Continuous or Discrete) or Categorical (Nominal, Ordinal, or Binary), **AND (ii)** identify which measurement scale each variable belongs to.

(c) [**2 marks**] Differentiate between the three levels of analytics: **Descriptive, Predictive, and Prescriptive**. For each level, state the core question it answers, give one example technique, and explain how these three levels represent an **increasing maturity** in an organization's analytical capability.

(d) [**2 marks**] Compare and contrast **Generative AI** and **Agentic AI** across at least four dimensions (e.g., focus, autonomy, memory, output type). Then explain, with an example, how Generative AI can serve as a *component* within an Agentic AI system rather than being a standalone paradigm.

(e) [**2 marks**] Explain the conceptual differences among **Data Mining, Machine Learning, Deep Learning, and Data Science**. How are these four fields related to each other? Use a brief hierarchical or Venn-diagram-style description showing which field encompasses which. Give one example task that belongs to each field.

Question 2 — AI Agents: Concepts, Architecture, Workflows and Best Practices [5 Marks]

(a) [**2 marks**] An AI Agent is fundamentally different from a simple chatbot. Describe the **six core components** of an AI Agent (Goal/Instruction, Model/Reasoning Engine, Memory/State, Tools & Actions, Policy & Guardrails, Feedback Loop). For each component, write one sentence explaining its role and what would happen if that component were missing or poorly designed.

(b) [**1.5 marks**] Compare **Single-Agent** and **Multi-Agent** systems. Under what circumstances would a multi-agent architecture be preferred over a single-agent design? Discuss at least two advantages and two risks of multi-agent systems, using a concrete educational or business scenario as an example.

(c) [**1.5 marks**] AI Agents introduce unique **risks** such as hallucinations, over-automation, tool misuse, and prompt injection. Choose any **two** of these risks and for each: **(i)** explain what the risk means in practical terms, **(ii)** describe a realistic scenario where it could cause harm, and **(iii)** recommend a specific design best practice from the lecture to mitigate it.

Question 3 — Association Rule Mining (ARM) [5 Marks]

(a) [**2 marks**] Consider the following transaction dataset for a campus bookshop:

TID	Items Purchased
T1	{Notebook, Pen, Highlighter}
T2	{Notebook, Pen}
T3	{Pen, Highlighter, Eraser}
T4	{Notebook, Highlighter, Eraser}
T5	{Notebook, Pen, Highlighter, Eraser}
T6	{Pen, Eraser}
T7	{Notebook, Pen, Eraser}
T8	{Highlighter, Eraser}

For the rule **{Notebook, Pen} → {Highlighter}**, calculate **Support, Confidence, Lift, and Conviction** by hand. Show all working steps. ($N = 8$)

(b) [**1.5 marks**] The ARM lecture demonstrated that a rule with the *highest confidence* is not necessarily the most useful — what matters more is **Lift**. Explain why relying solely on Confidence can be misleading. In your answer, describe what Lift measures, what a Lift value of exactly 1.0 means, and why a rule with Confidence = 0.80 but Lift = 1.0 should *not* be used for business decisions.

(c) [**1.5 marks**] Explain the **Apriori algorithm's** pruning strategy. What is the **anti-monotone (downward closure) property**, and how does it make the search for frequent itemsets computationally efficient? Illustrate with a brief example: if the itemset {A, B} is found to be *infrequent*, what can we immediately conclude about {A, B, C} and why?

Question 4 — Exploratory Data Analysis on mtcars [5 Marks]

(a) [**2 marks**] Explain the purpose and interpretation of three EDA visualization techniques — **histogram, boxplot, and scatter plot**. For each: **(i)** state what aspect of the data it reveals, **(ii)** describe one specific finding from the mtcars dataset that can be discovered using that visualization (referencing actual variables like `mpg`, `wt`, `hp`, or `cyl`), and **(iii)** explain how a data scientist would use that finding to guide subsequent modeling decisions.

(b) [**1.5 marks**] The correlation between `mpg` and `wt` in the mtcars dataset is $r = -0.87$, while the correlation between `mpg` and `drat` is $r = +0.68$. Interpret both correlation values in terms of **strength and direction**. Then explain: does the strong negative correlation between weight and MPG prove that *making a car lighter will improve its fuel efficiency*? Discuss the critical difference between **correlation and causation** in the context of EDA.

(c) [**1.5 marks**] The mtcars dataset contains 32 observations and 11 variables — all stored as numeric type in R. However, the lecture handout states that some variables (such as `cyl`, `vs`, `am`, `gear`) are actually **categorical**. Explain why correctly identifying variable types matters in EDA. What problems could arise if a data scientist treats `cyl` (with values 4, 6, 8) as a continuous numeric variable instead of a categorical one? Give an example of how the analysis or visualization would differ.

Question 5 — Predictive Modeling with Machine Learning [5 Marks]

(a) [**2 marks**] The predictive modeling lecture uses three evaluation metrics: **RMSE, MAE, and $\mathbf{R}^2$**. For each metric: **(i)** write the formula, **(ii)** explain what it measures in plain language, and **(iii)** state whether a higher or lower value indicates better performance. Then explain: why might RMSE and MAE give different rankings for two models? Specifically, under what conditions would RMSE penalize a model much more severely than MAE does?

(b) [**1.5 marks**] Explain the concept of **train/test split** and why it is essential in predictive modeling. In your answer, define **overfitting** and explain how it relates to the **bias-variance trade-off**. A student argues: *"I trained my Random Forest model on all 285 observations and got an R^2 of 0.98 — my model is excellent!"* Critique this claim. What mistake has the student made, and what would you recommend instead?

(c) [**1.5 marks**] Compare **Linear Regression** and **Random Forest** as predictive models. Discuss at least three dimensions of comparison (e.g., interpretability, handling non-linearity, overfitting behavior, speed, feature importance). In the context of predicting student performance from LMS engagement data, which model would you recommend for: **(i)** an early warning system where explainability to instructors matters, and **(ii)** a system where prediction accuracy is the top priority? Justify both choices.

END OF PAPER

Good Luck!